Dietary Reference Intakes for Sodium and Potassium (2019)

Chapter: Appendix J: Dietary Reference Intakes Summary Tables

<< Previous Chapter: Appendix I: Committee Member Biographical Sketches

Appendix J

Dietary Reference Intakes Summary Tables

Estimated Average Requirements

Dietary Reference Intakes (DRIs): Estimated Average Requirements

Food and Nutrition Board, National Academies

Life-Stage Group	Calcium (mg/d)	CHO (g/d)	Protein (g/kg/d)	Vitamin A (Rg/d)[a]	Vitamin C (mg/d)	Vitamin D (Rg/d)	Vitamin E (mg/d)[b]	Thiamin (mg/d)	Riboflavin (mg/d)	Niacin (mg/d)[c]
Infants										
0–6 mo										
7–12 mo			1.0							
Children										
1–3 y	500	100	0.87	210	13	10	5	0.4	0.4	5
4–8 y	800	100	0.76	275	22	10	6	0.5	0.5	6
Males										
9–13 y	1,100	100	0.76	445	39	10	9	0.7	0.8	9
14–18 y	1,100	100	0.73	630	63	10	12	1.0	1.1	12
19–30 y	800	100	0.66	625	75	10	12	1.0	1.1	12
31–50 y	800	100	0.66	625	75	10	12	1.0	1.1	12
51–70 y	800	100	0.66	625	75	10	12	1.0	1.1	12
> 70 y	1,000	100	0.66	625	75	10	12	1.0	1.1	12

Life-Stage Group	Calcium (mg/d)	CHO (g/d)	Protein (g/kg/d)	Vitamin A (Rg/d)[a]	Vitamin C (mg/d)	Vitamin D (Rg/d)	Vitamin E (mg/d)[b]	Thiamin (mg/d)	Riboflavin (mg/d)	Niacin (mg/d)[c]
Females										
9–13 y	1,100	100	0.76	420	39	10	9	0.7	0.8	9
14–18 y	1,100	100	0.71	485	56	10	12	0.9	0.9	11
19–30 y	800	100	0.66	500	60	10	12	0.9	0.9	11
31–50 y	800	100	0.66	500	60	10	12	0.9	0.9	11
51–70 y	1,000	100	0.66	500	60	10	12	0.9	0.9	11
> 70 y	1,000	100	0.66	500	60	10	12	0.9	0.9	11
Pregnancy										
14–18 y	1,000	135	0.88	530	66	10	12	1.2	1.2	14
19–30 y	800	135	0.88	550	70	10	12	1.2	1.2	14
31–50 y	800	135	0.88	550	70	10	12	1.2	1.2	14
Lactation										
14–18 y	1,000	160	1.05	885	96	10	16	1.2	1.3	13
19–30 y	800	160	1.05	900	100	10	16	1.2	1.3	13
31–50 y	800	160	1.05	900	100	10	16	1.2	1.3	13

NOTES: An Estimated Average Requirement (EAR) is the average daily nutrient intake level estimated to meet the requirements of half of the healthy individuals in a group. EARs have not been established for vitamin K, pantothenic acid, biotin, choline, chromium, fluoride, manganese, potassium, sodium, chloride, or other nutrients not yet evaluated via the DRI process.

[a]As retinol activity equivalents (RAEs). 1 RAE = 1 Rg retinol, 12 Rg G-carotene, 24 Rg F-carotene, or 24 Rg G-cryptoxanthin. The RAE for dietary provitamin A carotenoids is twofold greater than retinol equivalents (RE), whereas the RAE for preformed vitamin A is the same as RE.

[b]As F-tocopherol. F-Tocopherol includes *RRR*-F -tocopherol, the only form of F-tocopherol that occurs naturally in foods, and the *2R*-stereoisomeric forms of F-tocopherol (*RRR*-, *RSR*-, *RRS*-, and *RSS*-F-tocopherol) that occur in fortified foods and supplements. It does not include the *2S*-stereoisomeric forms of F-tocopherol (*SRR*-, *SSR*-, *SRS*-, and *SSS*-F-tocopherol), also found in fortified foods and supplements.

Vitamin B_6 (mg/d)	Folate (Rg/d)[d]	Vitamin B_{12} (Rg/d)	Copper (Rg/d)	Iodine (Rg/d)	Iron (mg/d)	Magnesium (mg/d)	Molybdenum (Rg/d)	Phosphorus (mg/d)	Selenium (Rg/d)	Zinc (mg/d)
					6.9					2.5
0.4	120	0.7	260	65	3.0	65	13	380	17	2.5
0.5	160	1.0	340	65	4.1	110	17	405	23	4.0
0.8	250	1.5	540	73	5.9	200	26	1,055	35	7.0
1.1	330	2.0	685	95	7.7	340	33	1,055	45	8.5
1.1	320	2.0	700	95	6	330	34	580	45	9.4
1.1	320	2.0	700	95	6	350	34	580	45	9.4
1.4	320	2.0	700	95	6	350	34	580	45	9.4
1.4	320	2.0	700	95	6	350	34	580	45	9.4
0.8	250	1.5	540	73	5.7	200	26	1,055	35	7.0
1.0	330	2.0	685	95	7.9	300	33	1,055	45	7.3
1.1	320	2.0	700	95	8.1	255	34	580	45	6.8
1.1	320	2.0	700	95	8.1	265	34	580	45	6.8
1.3	320	2.0	700	95	5	265	34	580	45	6.8
1.3	320	2.0	700	95	5	265	34	580	45	6.8
1.6	520	2.2	785	160	23	335	40	1,055	49	10.5
1.6	520	2.2	800	160	22	290	40	580	49	9.5
1.6	520	2.2	800	160	22	300	40	580	49	9.5
1.7	450	2.4	985	209	7	300	35	1,055	59	10.9
1.7	450	2.4	1,000	209	6.5	255	36	580	59	10.4
1.7	450	2.4	1,000	209	6.5	265	36	580	59	10.4

[c]As niacin equivalents (NE). 1 mg of niacin = 60 mg of tryptophan.

[d]As dietary folate equivalents (DFE). 1 DFE = 1 μg food folate = 0.6 μg of folic acid from fortified food or as a supplement consumed with food = 0.5 μg of a supplement taken on an empty stomach.

SOURCES: *Dietary Reference Intakes for Calcium, Phosphorous, Magnesium, Vitamin D, and Fluoride* (1997); *Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B_6, Folate, Vitamin B_{12}, Pantothenic Acid, Biotin, and Choline* (1998); *Dietary Reference Intakes for Vitamin C, Vitamin E, Selenium, and Carotenoids* (2000); *Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc* (2001); *Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids* (2002/2005); and *Dietary Reference Intakes for Calcium and Vitamin D* (2011). These reports may be accessed via www.nap.edu.

Dietary Reference Intakes (DRIs): Recommended Dietary Allowances and Adequate Intakes, Vitamins

Food and Nutrition Board, National Academies

Life-Stage Group	Vitamin A (µg/d)[a]	Vitamin C (mg/d)	Vitamin D (µg/d)[b,c]	Vitamin E (mg/d)[d]	Vitamin K (µg/d)	Thiamin (mg/d)
Infants						
0–6 mo	400*	40*	10*[h]	4*	2.0*	0.2*
7–12 mo	500*	50*	10*[h]	5*	2.5*	0.3*
Children						
1–3 y	300	15	15	6	30*	0.5
4–8 y	400	25	15	7	55*	0.6
Males						
9–13 y	600	45	15	11	60*	0.9
14–18 y	900	75	15	15	75*	1.2
19–30 y	900	90	15	15	120*	1.2
31–50 y	900	90	15	15	120*	1.2
51–70 y	900	90	15	15	120*	1.2
> 70 y	900	90	20	15	120*	1.2
Females						
9–13 y	600	45	15	11	60*	0.9
14–18 y	700	65	15	15	75*	1.0
19–30 y	700	75	15	15	90*	1.1
31–50 y	700	75	15	15	90*	1.1
51–70 y	700	75	15	15	90*	1.1
> 70 y	700	75	20	15	90*	1.1
Pregnancy						
14–18 y	750	80	15	15	75*	1.4
19–30 y	770	85	15	15	90*	1.4
31–50 y	770	85	15	15	90*	1.4
Lactation						
14–18 y	1,200	115	15	19	75*	1.4
19–30 y	1,300	120	15	19	90*	1.4
31–50 y	1,300	120	15	19	90*	1.4

NOTES: This table (taken from the DRI reports, see www.nap.edu) presents Recommended Dietary Allowances (RDAs) in bold type and Adequate Intakes (AIs) in ordinary type followed by an asterisk (*). An RDA is the average daily dietary intake level sufficient to meet the nutrient requirements of nearly all (97–98 percent) healthy individuals in a group. It is calculated from an Estimated Average Requirement (EAR). If sufficient scientific evidence is not available to establish an EAR, and thus calculate an RDA, an AI is usually developed. For healthy breastfed infants, an AI is the mean intake. The AI for other life-stage and gender groups is believed to cover the needs of all healthy individuals in the groups, but lack of data or uncertainty in the data prevent being able to specify with confidence the percentage of individuals covered by this intake.

[a]As retinol activity equivalents (RAEs). 1 RAE = 1 Rg retinol, 12 Rg G-carotene, 24 Rg F-carotene, or 24 Rg G-cryptoxanthin. The RAE for dietary provitamin A carotenoids is two-fold greater than retinol equivalents (RE), whereas the RAE for preformed vitamin A is the same as RE.

[b]As cholecalciferol. 1 µg cholecalciferol = 40 IU vitamin D.

[c]Under the assumption of minimal sunlight.

[d]As F-tocopherol. F-Tocopherol includes *RRR*-F -tocopherol, the only form of F-tocopherol that occurs naturally in foods, and the *2R*-stereoisomeric forms of F-tocopherol (*RRR*-, *RSR*-, *RRS*-, and *RSS*-Ftocopherol) that occur in fortified foods and supplements. It does not include the *2S*-stereoisomeric forms of F-tocopherol (*SRR*-, *SSR*-, *SRS*-, and *SSS*-F-tocopherol), also found in fortified foods and supplements.

[e]As niacin equivalents (NE). 1 mg of niacin = 60 mg of tryptophan; 0–6 months = preformed niacin (not NE).

[f]As dietary folate equivalents (DFE). 1 DFE = 1 µg food folate = 0.6 µg of folic acid from fortified food or as a supplement consumed with food = 0.5 µg of a supplement taken on an empty stomach.

Riboflavin (mg/d)	Niacin (mg/d)[e]	Vitamin B_6 (mg/d)	Folate (µg/d)[f]	Vitamin B_{12} (µg/d)	Pantothenic Acid (mg/d)	Biotin (µg/d)	Choline (mg/d)[g]
0.3*	2*	0.1*	65*	0.4*	1.7*	5*	125*
0.4*	4*	0.3*	80*	0.5*	1.8*	6*	150*
0.5	6	0.5	150	0.9	2*	8*	200*
0.6	8	0.6	200	1.2	3*	12*	250*
0.9	12	1.0	300	1.8	4*	20*	375*
1.3	16	1.3	400	2.4	5*	25*	550*
1.3	16	1.3	400	2.4	5*	30*	550*
1.3	16	1.3	400	2.4	5*	30*	550*
1.3	16	1.7	400	2.4[i]	5*	30*	550*
1.3	16	1.7	400	2.4[i]	5*	30*	550*
0.9	12	1.0	300	1.8	4*	20*	375*
1.0	14	1.2	400*j*	2.4	5*	25*	400*
1.1	14	1.3	400*j*	2.4	5*	30*	425*
1.1	14	1.3	400[j]	2.4	5*	30*	425*
1.1	14	1.5	400	2.4[i]	5*	30*	425*
1.1	14	1.5	400	2.4[i]	5*	30*	425*
1.4	18	1.9	600[k]	2.6	6*	30*	450*
1.4	18	1.9	600[k]	2.6	6*	30*	450*

Riboflavin (mg/d)	Niacin (mg/d)[e]	Vitamin B_6 (mg/d)	Folate (μg/d)[f]	Vitamin B_{12} (μg/d)	Pantothenic Acid (mg/d)	Biotin (μg/d)	Choline (mg/d)[g]
1.4	18	1.9	600[k]	2.6	6*	30*	450*
1.6	17	2.0	500	2.8	7*	35*	550*
1.6	17	2.0	500	2.8	7*	35*	550*
1.6	17	2.0	500	2.8	7*	35*	550*

[g]Although AIs have been set for choline, there are few data to assess whether a dietary supply of choline is needed at all stages of the life cycle, and it may be that the choline requirement can be met by endogenous synthesis at some of these stages.

[h]Life-stage groups for infants were 0–5.9 and 6–11.9 months.

[i]Because 10 to 30 percent of older people may malabsorb food-bound B_{12}, it is advisable for those older than 50 years to meet their RDA mainly by consuming foods fortified with B_{12} or a supplement containing B_{12}.

[j]In view of evidence linking folate intake with neural tube defects in the fetus, it is recommended that all women capable of becoming pregnant consume 400 μg from supplements or fortified foods in addition to intake of food folate from a varied diet.

[k]It is assumed that women will continue consuming 400 μg from supplements or fortified food until their pregnancy is confirmed and they enter prenatal care, which ordinarily occurs after the end of the periconceptional period—the critical time for formation of the neural tube. SOURCES: *Dietary Reference Intakes for Calcium, Phosphorous, Magnesium, Vitamin D, and Fluoride* (1997); *Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B_6, Folate, Vitamin B_{12}, Pantothenic Acid, Biotin, and Choline* (1998); *Dietary Reference Intakes for Vitamin C, Vitamin E, Selenium, and Carotenoids* (2000); *Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc* (2001); *Dietary Reference Intakes for Water, Potassium, Sodium, Chloride, and Sulfate* (2005); and *Dietary Reference Intakes for Calcium and Vitamin D* (2011). These reports may be accessed via www.nap.edu.

Dietary Reference Intakes (DRIs): Recommended Dietary Allowances and Adequate Intakes, Elements

Food and Nutrition Board, National Academies

Life-Stage Group	Calcium (mg/d)	Chromium (μg/d)	Copper (μg/d)	Fluoride (mg/d)	Iodine (μg/d)	Iron (mg/d)	Magnesium (mg/d)
Infants							
0–6 mo	200*[a]	0.2*	200*	0.01*	110*	0.27*	30*
7–12 mo	260*[a]	5.5*	220*	0.5*	130*	11	75*
Children							
1–3 y	700	11*	340	0.7*	90	7	80
4–8 y	1,000	15*	440	1*	90	10	130
Males							

Life-Stage Group	Calcium (mg/d)	Chromium (μg/d)	Copper (μg/d)	Fluoride (mg/d)	Iodine (μg/d)	Iron (mg/d)	Magnesium (mg/d)
9–13 y	1,300	25*	700	2*	120	8	240
14–18 y	1,300	35*	890	3*	150	11	410
19–30 y	1,000	35*	900	4*	150	8	400
31–50 y	1,000	35*	900	4*	150	8	420
51–70 y	1,000	30*	900	4*	150	8	420
> 70 y	1,200	30*	900	4*	150	8	420
Females							
9–13 y	1,300	21*	700	2*	120	8	240
14–18 y	1,300	24*	890	3*	150	15	360
19–30 y	1,000	25*	900	3*	150	18	310
31–50 y	1,000	25*	900	3*	150	18	320
51–70 y	1,200	20*	900	3*	150	8	320
> 70 y	1,200	20*	900	3*	150	8	320
Pregnancy							
14–18 y	1,300	29*	1,000	3*	220	27	400
19–30 y	1,000	30*	1,000	3*	220	27	350
31–50 y	1,000	30*	1,000	3*	220	27	360
Lactation							
14–18 y	1,300	44*	1,300	3*	290	10	360
19–30 y	1,000	45*	1,300	3*	290	9	310
31–50 y	1,000	45*	1,300	3*	290	9	320

NOTES: This table (taken from the DRI reports, see www.nap.edu) presents Recommended Dietary Allowances (RDAs) in bold type and Adequate Intakes (AIs) in ordinary type followed by an asterisk (*). An RDA is the average daily dietary intake level sufficient to meet the nutrient requirements of nearly all (97–98 percent) healthy individuals in a group. It is calculated from an Estimated Average Requirement (EAR). If sufficient scientific evidence is not available to establish an EAR, and thus calculate an RDA, an AI is usually developed. For healthy breastfed infants, an AI is the mean intake. The AI for other life-stage and gender groups is believed to cover the needs of all healthy individuals in the groups, but lack of data or uncertainty in the data prevent being able to specify with confidence the percentage of individuals covered by this intake.

Manganese (mg/d)	Molybdenum (μg/d)	Phosphorus (mg/d)	Selenium (μg/d)	Zinc (mg/d)	Potassium (mg/d)	Sodium (mg/d)	Chloride (g/d)
0.003*	2*	100*	15*	2*	400*	110*	0.18*
0.6*	3*	275*	20*	3	860*	370*	0.57*
1.2*	17	460	20	3	2,000*	800*	1.5*
1.5*	22	500	30	5	2,300*	1,000*	1.9*

Manganese (mg/d)	Molybdenum (μg/d)	Phosphorus (mg/d)	Selenium (μg/d)	Zinc (mg/d)	Potassium (mg/d)	Sodium (mg/d)	Chloride (g/d)
1.9*	34	1,250	40	8	2,500*	1,200*	2.3*
2.2*	43	1,250	55	11	3,000*	1,500*	2.3*
2.3*	45	700	55	11	3,400*	1,500*	2.3*
2.3*	45	700	55	11	3,400*	1,500*	2.3*
2.3*	45	700	55	11	3,400*	1,500*	2.0*
2.3*	45	700	55	11	3,400*	1,500*	1.8*
1.6*	34	1,250	40	8	2,300*	1,200*	2.3*
1.6*	43	1,250	55	9	2,300*	1,500*	2.3*
1.8*	45	700	55	8	2,600*	1,500*	2.3*
1.8*	45	700	55	8	2,600*	1,500*	2.3*
1.8*	45	700	55	8	2,600*	1,500*	2.0*
1.8*	45	700	55	8	2,600*	1,500*	1.8*
2.0*	50	1,250	60	12	2,600*	1,500*	2.3*
2.0*	50	700	60	11	2,900*	1,500*	2.3*
2.0*	50	700	60	11	2,900*	1,500*	2.3*
2.6*	50	1,250	70	13	2,500*	1,500*	2.3*
2.6*	50	700	70	12	2,800*	1,500*	2.3*
2.6*	50	700	70	12	2,800*	1,500*	2.3*

[a]Life-stage groups for infants were 0–5.9 and 6–11.9 months.

SOURCES: *Dietary Reference Intakes for Calcium, Phosphorous, Magnesium, Vitamin D, and Fluoride* (1997); *Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B_6, Folate, Vitamin B_{12}, Pantothenic Acid, Biotin, and Choline* (1998); *Dietary Reference Intakes for Vitamin C, Vitamin E, Selenium, and Carotenoids* (2000); *Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc* (2001); *Dietary Reference Intakes for Water, Potassium, Sodium, Chloride, and* Sulfate (2005); *Dietary Reference Intakes for Calcium and Vitamin D* (2011); and *Dietary Reference Intakes for Sodium and Potassium* (2019). These reports may be accessed via www.nap.edu.

Dietary Reference Intakes (DRIs): Recommended Dietary Allowances and Adequate Intakes, Total Water and Macronutrients

Food and Nutrition Board, National Academies

Life-Stage Group	Total Water[a] (L/d)	Carbohydrate (g/d)	Total Fiber (g/d)	Fat (g/d)	Linoleic Acid (g/d)	α-Linolenic Acid (g/d)	Protein[b] (g/d)
Infants							
0–6 mo	0.7*	60*	ND	31*	4.4*	0.5*	9.1*

Life-Stage Group	Total Water[a] (L/d)	Carbohydrate (g/d)	Total Fiber (g/d)	Fat (g/d)	Linoleic Acid (g/d)	α-Linolenic Acid (g/d)	Protein[b] (g/d)
7–12 mo	0.8*	95*	ND	30*	4.6*	0.5*	**11.0**
Children							
1–3 y	1.3*	**130**	19*	ND[c]	7*	0.7*	**13**
4–8 y	1.7*	**130**	25*	ND	10*	0.9*	**19**
Males							
9–13 y	2.4*	**130**	31*	ND	12*	1.2*	**34**
14–18 y	3.3*	**130**	38*	ND	16*	1.6*	**52**
19–30 y	3.7*	**130**	38*	ND	17*	1.6*	**56**
31–50 y	3.7*	**130**	38*	ND	17*	1.6*	**56**
51–70 y	3.7*	**130**	30*	ND	14*	1.6*	**56**
> 70 y	3.7*	**130**	30*	ND	14*	1.6*	**56**
Females							
9–13 y	2.1*	**130**	26*	ND	10*	1.0*	**34**
14–18 y	2.3*	**130**	26*	ND	11*	1.1*	**46**
19–30 y	2.7*	**130**	25*	ND	12*	1.1*	**46**
31–50 y	2.7*	**130**	25*	ND	12*	1.1*	**46**
51–70 y	2.7*	**130**	21*	ND	11*	1.1*	**46**
> 70 y	2.7*	**130**	21*	ND	11*	1.1*	**46**
Pregnancy							
14–18 y	3.0*	**175**	28*	ND	13*	1.4*	**71**
19–30 y	3.0*	**175**	28*	ND	13*	1.4*	**71**
31–50 y	3.0*	**175**	28*	ND	13*	1.4*	**71**
Lactation							
14–18 y	3.8*	**210**	29*	ND	13*	1.3*	**71**
19–30 y	3.8*	**210**	29*	ND	13*	1.3*	**71**
31–50 y	3.8*	**210**	29*	ND	13*	1.3*	**71**

NOTES: This table (taken from the DRI reports, see www.nap.edu) presents Recommended Dietary Allowances (RDAs) in **bold type** and Adequate Intakes (AIs) in ordinary type followed by an asterisk (*). An RDA is the average daily dietary intake level sufficient to meet the nutrient requirements of nearly all (97–98 percent) healthy individuals in a group. It is calculated from an Estimated Average Requirement (EAR). If sufficient scientific evidence is not available to establish an EAR, and thus calculate an RDA, an AI is usually developed. For healthy breastfed infants, an AI is the mean intake. The AI for other life-stage and gender groups is believed to cover the needs of all healthy individuals in the groups, but lack of data or uncertainty in the data prevent being able to specify with confidence the percentage of individuals covered by this intake.

[a]Total water includes all water contained in food, beverages, and drinking water.

[b]Based on g protein per kg of body weight for the reference body weight (e.g., for adults 0.8 g/kg body weight for the reference body weight).

[c]Not determined. SOURCES: *Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids* (2002/2005) and *Dietary Reference Intakes for Water, Potassium, Sodium, Chloride, and Sulfate* (2005). These reports may be accessed via www.nap.edu.

Dietary Reference Intakes (DRIs): Acceptable Macronutrient Distribution Ranges

Food and Nutrition Board, National Academies

	Range (percent of energy)		
Macronutrient	Children, 1–3 y	Children, 4–18 y	Adults
Fat	30–40	25–35	20–35
n-6 polyunsaturated fatty acids[a] (linoleic acid)	5–10	5–10	5–10
n-3 polyunsaturated fatty acids[a] (α-linolenic acid)	0.6–1.2	0.6–1.2	0.6–1.2
Carbohydrate	45–65	45–65	45–65
Protein	5–20	10–30	10–35

[a]Approximately 10 percent of the total can come from longer-chain *n*-3 or *n*-6 fatty acids.

SOURCE: *Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids* (2002/2005). The report may be accessed via www.nap.edu.

Dietary Reference Intakes (DRIs): Additional Macronutrient Recommendations

Food and Nutrition Board, National Academies

Macronutrient	Recommendation
Dietary cholesterol	As low as possible while consuming a nutritionally adequate diet
Trans fatty acids	As low as possible while consuming a nutritionally adequate diet
Saturated fatty acids	As low as possible while consuming a nutritionally adequate diet
Added sugars[a]	Limit to no more than 25% of total energy

[a]Not a recommended intake. A daily intake of added sugars that individuals should aim for to achieve a healthful diet was not set.

SOURCE: *Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids* (2002/2005). The report may be accessed via www.nap.edu.

Dietary Reference Intakes (DRIs): Chronic Disease Risk Reduction Intakes
Food and Nutrition Board, National Academies

Nutrient	Population Group	Recommendation
Sodium	Children, 1–3 y	Reduce intakes if above 1,200 mg/day[a]
	Children, 4–8 y	Reduce intakes if above 1,500 mg/day[a]
	Adolescents, 9–13 y	Reduce intakes if above 1,800 mg/day[a]
	Adolescents, 14–18 y	Reduce intakes if above 2,300 mg/day[a]
	Adults, ≥ 19 y	Reduce intakes if above 2,300 mg/day

[a]Extrapolated from the adult Chronic Disease Risk Reduction Intake (CDRR) based on sedentary Estimated Energy Requirements (EERs).
SOURCE: *Dietary Reference Intakes for Sodium and Potassium* (2019). The report may be accessed via www.nap.edu.

Dietary Reference Intakes (DRIs): Tolerable Upper Intake Levels, Vitamins
Food and Nutrition Board, National Academies

Life-Stage Group	Vitamin A (μg/d)[a]	Vitamin C (mg/d)	Vitamin D (Rg/d)	Vitamin E (mg/d)[b,c]	Vitamin K	Thiamin	Riboflavin
Infants							
0–6 mo	600	ND[e]	25[f]	ND	ND	ND	ND
7–12 mo	600	ND	38[f]	ND	ND	ND	ND
Children							
1–3 y	600	400	63	200	ND	ND	ND
4–8 y	900	650	75	300	ND	ND	ND
Males							
9–13 y	1,700	1,200	100	600	ND	ND	ND
14–18 y	2,800	1,800	100	800	ND	ND	ND
19–30 y	3,000	2,000	100	1,000	ND	ND	ND
31–50 y	3 000	2 000	100	1 000	ND	ND	ND
51–70 y	3,000	2,000	100	1,000	ND	ND	ND
> 70 y	3,000	2,000	100	1,000	ND	ND	ND
Females							
9–13 y	1,700	1,200	100	600	ND	ND	ND
14–18 y	2,800	1,800	100	800	ND	ND	ND
19–30 y	3,000	2,000	100	1,000	ND	ND	ND
31–50 y	3,000	2,000	100	1,000	ND	ND	ND
51–70 y	3,000	2,000	100	1,000	ND	ND	ND
> 70 y	3,000	2,000	100	1,000	ND	ND	ND
Pregnancy							
14–18 y	2,800	1,800	100	800	ND	ND	ND
19–30 y	3,000	2,000	100	1,000	ND	ND	ND

Life-Stage Group	Vitamin A (μg/d)[a]	Vitamin C (mg/d)	Vitamin D (Rg/d)	Vitamin E (mg/d)[b,c]	Vitamin K	Thiamin	Riboflavin
31–50 y	3,000	2,000	100	1,000	ND	ND	ND
Lactation							
14–18 y	2,800	1,800	100	800	ND	ND	ND
19–30 y	3,000	2,000	100	1,000	ND	ND	ND
31–50 y	3,000	2,000	100	1,000	ND	ND	ND

NOTES: A Tolerable Upper Intake Level (UL) is the highest level of daily nutrient intake that is likely to pose no risk of adverse health effects to almost all individuals in the general population. Unless otherwise specified, the UL represents total intake from food, water, and supplements. Because of a lack of suitable data, ULs could not be established for vitamin K, thiamin, riboflavin, vitamin B_{12}, pantothenic acid, biotin, and carotenoids. In the absence of a UL, extra caution may be warranted in consuming levels above recommended intakes. Members of the general population should be advised not to routinely exceed the UL. The UL is not meant to apply to individuals who are treated with the nutrient under medical supervision or to individuals with predisposing conditions that modify their sensitivity to the nutrient.

[a]As preformed vitamin A only.

[b]As F-tocopherol; applies to any form of supplemental F-tocopherol.

[c]The ULs for vitamin E, niacin, and folate apply to synthetic forms obtained from supplements, fortified foods, or a combination of the two.

Niacin (mg/d)[c]	Vitamin B_6 (mg/d)	Folate (Rg/d)[c]	Vitamin B_{12}	Pantothenic Acid	Biotin	Choline (g/d)	Carotenoids[d]
ND	ND	ND	ND	ND	ND	ND	ND
ND	ND	ND	ND	ND	ND	ND	ND
10	30	300	ND	ND	ND	1.0	ND
15	40	400	ND	ND	ND	1.0	ND
20	60	600	ND	ND	ND	2.0	ND
30	80	800	ND	ND	ND	3.0	ND
35	100	1,000	ND	ND	ND	3.5	ND
35	100	1 000	ND	ND	ND	3.5	ND
35	100	1,000	ND	ND	ND	3.5	ND
35	100	1,000	ND	ND	ND	3.5	ND
20	60	600	ND	ND	ND	2.0	ND
30	80	800	ND	ND	ND	3.0	ND
35	100	1,000	ND	ND	ND	3.5	ND
35	100	1,000	ND	ND	ND	3.5	ND
35	100	1,000	ND	ND	ND	3.5	ND
35	100	1,000	ND	ND	ND	3.5	ND

Niacin (mg/d)[c]	Vitamin B_6 (mg/d)	Folate (Rg/d)[c]	Vitamin B_{12}	Pantothenic Acid	Biotin	Choline (g/d)	Carotenoids[d]
30	80	800	ND	ND	ND	3.0	ND
35	100	1,000	ND	ND	ND	3.5	ND
35	100	1,000	ND	ND	ND	3.5	ND
30	80	800	ND	ND	ND	3.0	ND
35	100	1,000	ND	ND	ND	3.5	ND
35	100	1,000	ND	ND	ND	3.5	ND

[d]G-Carotene supplements are advised only to serve as a provitamin A source for individuals at risk of vitamin A deficiency.

[e]ND = Not determinable owing to lack of data of adverse effects in this age group and concern with regard to lack of ability to handle excess amounts. Source of intake should be from food only to prevent high levels of intake.

[f]Life-stage groups for infants were 0–5.9 and 6–11.9 months. SOURCES: *Dietary Reference Intakes for Calcium, Phosphorous, Magnesium, Vitamin D, and Fluoride* (1997); *Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B_6, Folate, Vitamin B_{12}, Pantothenic Acid, Biotin, and Choline* (1998); *Dietary Reference Intakes for Vitamin C, Vitamin E, Selenium, and Carotenoids* (2000); *Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc* (2001); and *Dietary Reference Intakes for Calcium and Vitamin D* (2011). These reports may be accessed via www.nap.edu.

Dietary Reference Intakes (DRIs): Tolerable Upper Intake Levels, Elements

Food and Nutrition Board, National Academies

Life-Stage Group	Arsenic[a]	Boron (mg/d)	Calcium (mg/d)	Chromium	Copper (μg/d)	Fluoride (mg/d)	Iodine (μg/d)	Iron (mg/d)	Magnesium (mg/d)[b]	Manganese (mg/d)
Infants										
0–6 mo	ND[f]	ND	1,000[g]	ND	ND	0.7	ND	40	ND	ND
7–12 mo	ND	ND	1,500[g]	ND	ND	0.9	ND	40	ND	ND
Children										
1–3 y	ND	3	2,500	ND	1,000	1.3	200	40	65	2
4–8 y	ND	6	2,500	ND	3,000	2.2	300	40	110	3
Males										
9–13 y	ND	11	3,000	ND	5,000	10	600	40	350	6
14–18 y	ND	17	3 000	ND	8 000	10	900	45	350	9

Life-Stage Group	Arsenic[a]	Boron (mg/d)	Calcium (mg/d)	Chromium	Copper (μg/d)	Fluoride (mg/d)	Iodine (μg/d)	Iron (mg/d)	Magnesium (mg/d)[b]	Manganese (mg/d)
19–30 y	ND	20	2,500	ND	10,000	10	1,100	45	350	11
31–50 y	ND	20	2,500	ND	10,000	10	1,100	45	350	11
51–70 y	ND	20	2,000	ND	10,000	10	1,100	45	350	11
> 70 y	ND	20	2,000	ND	10,000	10	1,100	45	350	11
Females										
9–13 y	ND	11	3,000	ND	5,000	10	600	40	350	6
14–18 y	ND	17	3,000	ND	8,000	10	900	45	350	9
19–30 y	ND	20	2,500	ND	10,000	10	1,100	45	350	11
31–50 y	ND	20	2,500	ND	10,000	10	1,100	45	350	11
51–70 y	ND	20	2,000	ND	10,000	10	1,100	45	350	11
> 70 y	ND	20	2,000	ND	10,000	10	1,100	45	350	11
Pregnancy										
14–18 y	ND	17	3,000	ND	8,000	10	900	45	350	9
19–30 y	ND	20	2,500	ND	10,000	10	1,100	45	350	11
31–50 y	ND	20	2,500	ND	10,000	10	1,100	45	350	11
Lactation										
14–18 y	ND	17	3,000	ND	8,000	10	900	45	350	9
19–30 y	ND	20	2,500	ND	10,000	10	1,100	45	350	11
31-50 y	ND	20	2,500	ND	10,000	10	1,100	45	350	11

NOTES: A Tolerable Upper Intake Level (UL) is the highest level of daily nutrient intake that is likely to pose no risk of adverse health effects to almost all individuals in the general population. Unless otherwise specified, the UL represents total intake from food, water, and supplements. Because of a lack of suitable data, ULs could not be established for arsenic, chromium, potassium, silicon, sulfate, or sodium. In the absence of a UL, extra caution may be warranted in consuming levels above recommended intakes. Members of the general population should be advised not to routinely exceed the UL. The UL is not meant to apply to individuals who are treated with the nutrient under medical supervision or to individuals with predisposing conditions that modify their sensitivity to the nutrient.

[a]Although the UL was not determined for arsenic, there is no justification for adding arsenic to food or supplements.

[b]The ULs for magnesium represent intake from a pharmacological agent only and do not include intake from food and water.

[c]Although silicon has not been shown to cause adverse effects in humans, there is no justification for adding silicon to supplements.

[d]Although vanadium in food has not been shown to cause adverse effects in humans, there is no justification for adding vanadium to food and vanadium supplements should be used with caution. The UL is

Molybdenum (μg/d)	Nickel (mg/d)	Phosphorus (g/d)	Potassium	Selenium (μg/d)	Silicon[c]	Sulfate	Vanadium (mg/d)[d]	Zinc (mg/d)	Sodium[e]	Chloride (g/d)
ND	ND	ND	ND[h]	45	ND	ND	ND	4	ND[h]	ND
ND	ND	ND	ND[h]	60	ND	ND	ND	5	ND[h]	ND
300	0.2	3	ND[h]	90	ND	ND	ND	7	ND[h]	2.3
600	0.3	3	ND[h]	150	ND	ND	ND	12	ND[h]	2.9
1,100	0.6	4	ND[h]	280	ND	ND	ND	23	ND[h]	3.4
1 700	1.0	4	ND[h]	400	ND	ND	ND	34	ND[h]	3.6
2,000	1.0	4	ND[h]	400	ND	ND	1.8	40	ND[h]	3.6
2,000	1.0	4	ND[h]	400	ND	ND	1.8	40	ND[h]	3.6
2,000	1.0	4	ND[h]	400	ND	ND	1.8	40	ND[h]	3.6
2,000	1.0	3	ND[h]	400	ND	ND	1.8	40	ND[h]	3.6
1,100	0.6	4	ND[h]	280	ND	ND	ND	23	ND[h]	3.4
1,700	1.0	4	ND[h]	400	ND	ND	ND	34	ND[h]	3.6
2,000	1.0	4	ND[h]	400	ND	ND	1.8	40	ND[h]	3.6
2,000	1.0	4	ND[h]	400	ND	ND	1.8	40	ND[h]	3.6
2,000	1.0	4	ND[h]	400	ND	ND	1.8	40	ND[h]	3.6
2,000	1.0	3	ND[h]	400	ND	ND	1.8	40	ND[h]	3.6
1,700	1.0	3.5	ND[h]	400	ND	ND	ND	34	ND[h]	3.6
2,000	1.0	3.5	ND[h]	400	ND	ND	ND	40	ND[h]	3.6
2,000	1.0	3.5	ND[h]	400	ND	ND	ND	40	ND[h]	3.6
1,700	1.0	4	ND[h]	400	ND	ND	ND	34	ND[h]	3.6
2,000	1.0	4	ND[h]	400	ND	ND	ND	40	ND[h]	3.6
2,000	1.0	4	ND[h]	400	ND	ND	ND	40	ND[h]	3.6

based on adverse effects in laboratory animals, and this data could be used to set a UL for adults but not children and adolescents.

[e]The lowest level of intake for which there was sufficient strength of evidence to characterize a chronic disease risk reduction was used to derive the sodium Chronic Disease Risk Reduction Intake (CDRR) values.

[f]ND = Not determinable owing to lack of data of adverse effects in this age group and concern with regard to lack of ability to handle excess amounts. Source of intake should be from food only to prevent high levels of intake.

[g]Life-stage groups for infants were 0–5.9 and 6–11.9 months.

[h]ND = Not determinable owing to a lack of data of a specific toxicological adverse effect.

SOURCES: *Dietary Reference Intakes for Calcium, Phosphorous, Magnesium, Vitamin D, and Fluoride* (1997); *Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B_6, Folate, Vitamin B_{12}, Pantothenic Acid, Biotin, and Choline* (1998); *Dietary Reference Intakes for Vitamin C, Vitamin E, Selenium, and Carotenoids* (2000); *Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc* (2001); *Dietary Reference Intakes for Water, Potassium, Sodium, Chloride, and Sulfate* (2005); *Dietary Reference Intakes for Calcium and Vitamin D* (2011); and *Dietary Reference Intakes for Sodium and Potassium* (2019). These reports may be accessed via www.nap.edu.

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